

Project Initialization and Planning Phase

Date	15 March 2024
Team ID	PNT2022TMID124356
Project Title	Opti crop AI based Recommendation System
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed **OptiCrop – AI-Based Crop Recommendation System** aims to help farmers select the most suitable crop using Machine Learning. The system analyzes soil nutrients and environmental conditions to recommend the best crop for cultivation, improving productivity and supporting better farming decisions

Project Overview	
Objective	Develop a Machine Learning-based crop recommendation system that predicts the most suitable crop using soil nutrients and environmental conditions through a simple Flask web application
Scope	The project focuses on preprocessing agricultural data, training a Machine Learning model, integrating it with Flask, and providing real-time crop recommendations through a user-friendly web interface
Problem Statement	
Description	Farmers often face difficulties in selecting the right crop due to changing soil nutrients and climate conditions, resulting in reduced crop yield and financial losses.
Impact	Incorrect crop selection leads to lower agricultural productivity, inefficient use of resources, and reduced profitability for farmers.
Proposed Solution	
Approach	Develop a Machine Learning model using Scikit-learn, integrate it with a Flask backend, preprocess applicant information, and generate loan approval predictions through a responsive web application.
Key Features	• AI-based crop recommendation system. • Real-time crop prediction using Flask. • Simple and user-friendly interface.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU specifications	Intel Core i5 or above GPU
Memory	RAM specifications	8 GB
Storage	Space for dataset, models and project files.	10 GB Free Disk Space
Software		
Frameworks	Backend Framework	Flask
Libraries	Machine Learning Libraries.	scikit-learn, pandas, Numpy, Joblib
Development Environment	IDE	Visual Studio Code
Data		
Data	Crop Recommendation Dataset	Kaggle Crop Recommendation Dataset (CSV Format, 2200 Records)